

Authors :

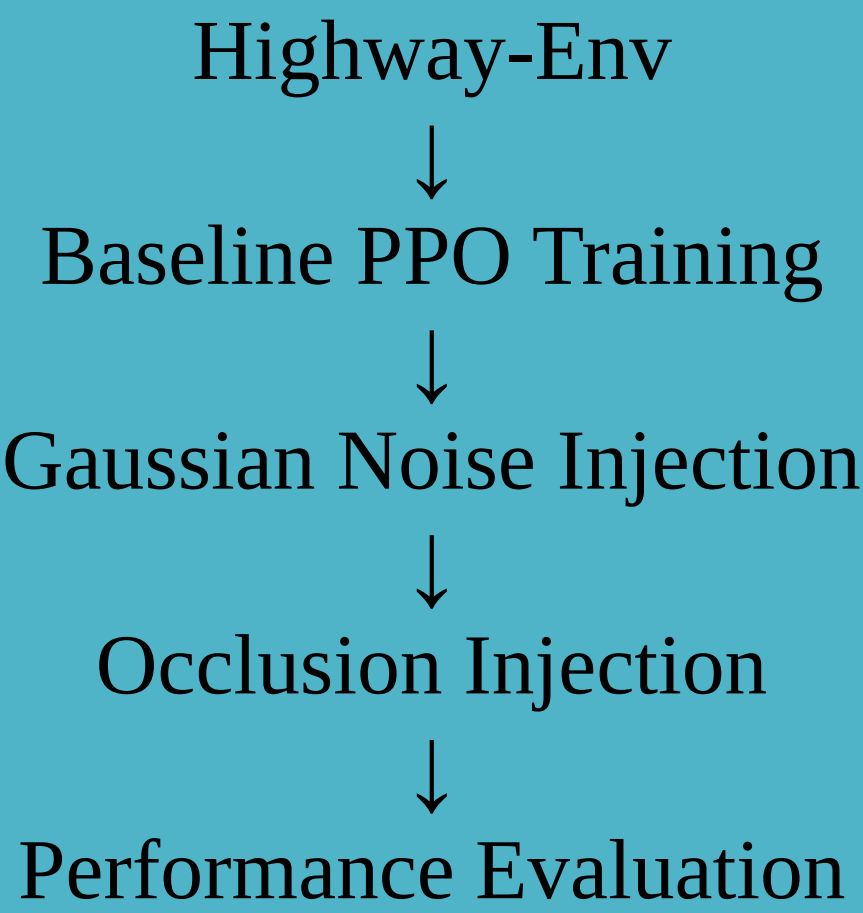
Basetti Sai Viswas
Dhanalakota Satya Ganga Sri Govinda Nandini
Kanika Yadav

Improving Robustness of Reinforcement Learning for Autonomous Driving under Sensor Noise and Partial Observability

Motivation

Autonomous vehicles rely heavily on sensor observations to perceive their surroundings. In real-world deployments, sensor measurements are often affected by noise, environmental conditions, and partial occlusions caused by surrounding vehicles or obstacles. These imperfections can significantly impact decision-making and driving safety. This work investigates the robustness of reinforcement learning policies under observation uncertainty by evaluating Proximal Policy Optimization (PPO) and Recurrent PPO in a simulated autonomous driving environment. The objective is to understand how well these agents maintain performance when exposed to increasing levels of sensor noise and partial observability.

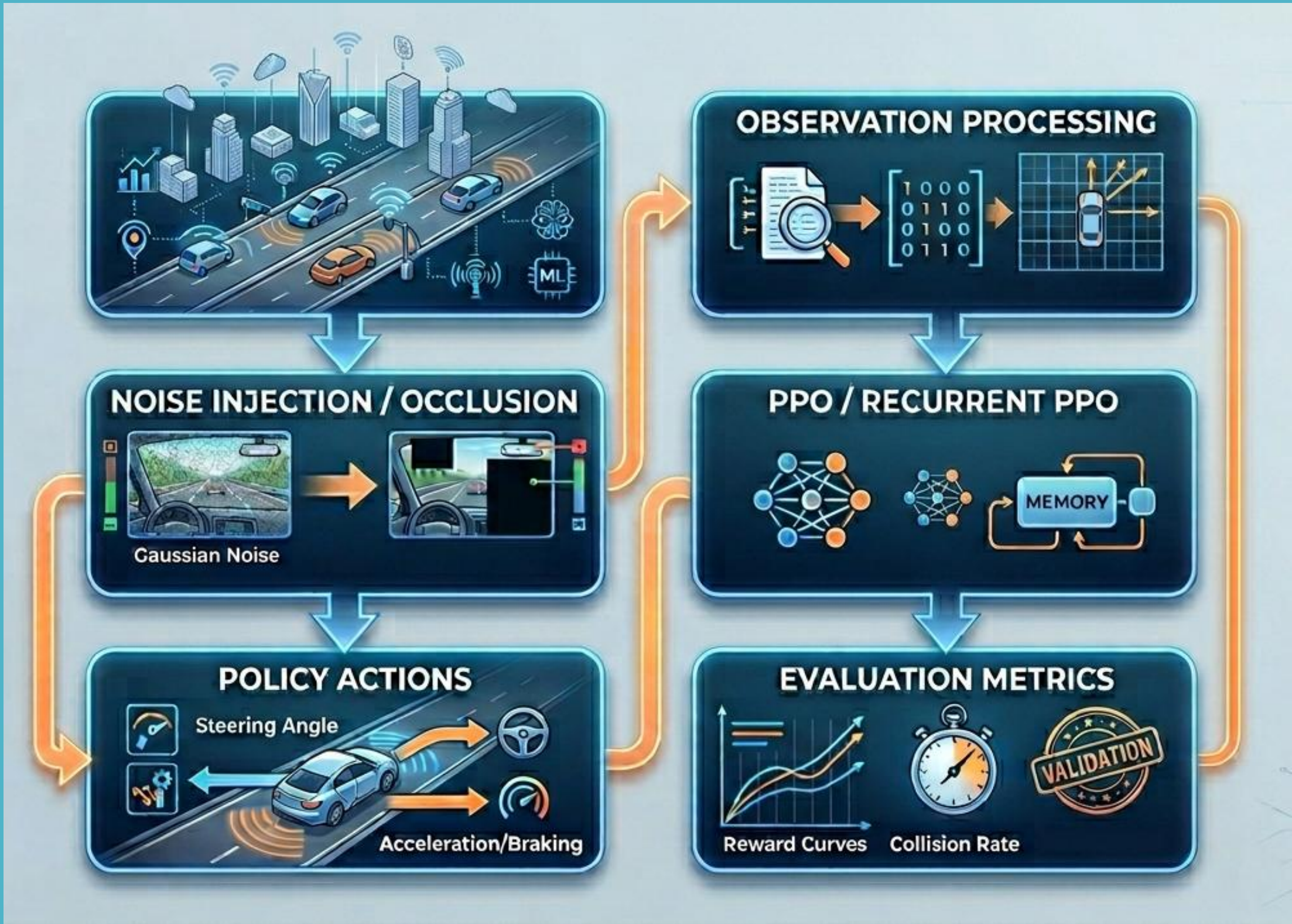
Methodology



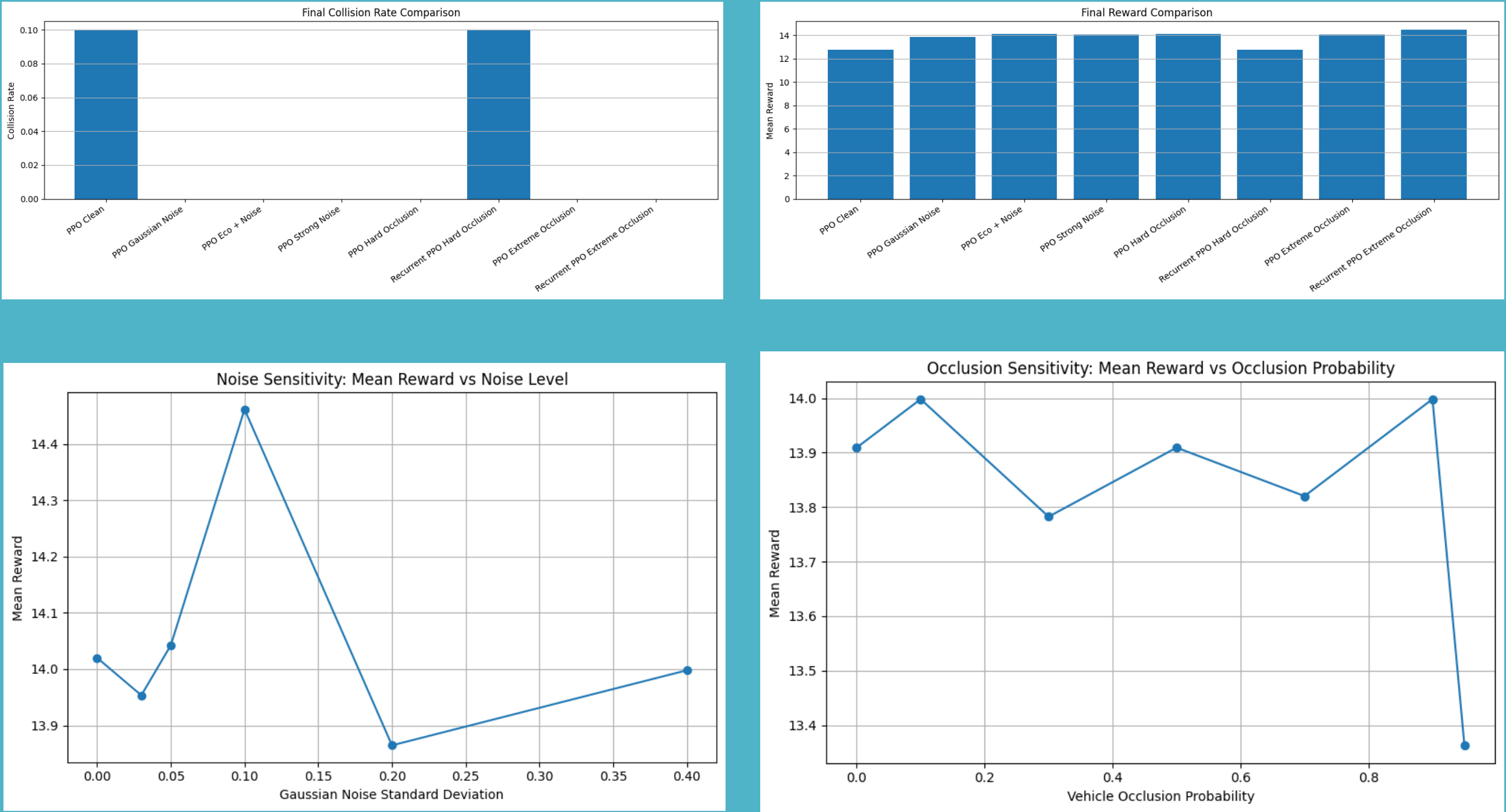
Environment



Pipeline Diagram



Results



Results Table

TABLE I
FINAL COMPARISON OF PPO AND RECURRENT PPO UNDER DIFFERENT CONDITIONS.

Experiment	Reward	Collision	Speed
PPO Clean 10k	12.75	0.10	19.99
PPO Gaussian Noise 10k	13.84	0.00	20.04
PPO Eco + Noise 10k	14.11	0.00	20.04
PPO Strong Noise 10k	14.06	0.00	20.04
PPO Extreme Occlusion 20k	14.06	0.00	20.04
Recurrent PPO Extreme 20k	12.36	0.70	23.71
PPO Extreme Occlusion 100k	14.00	0.05	20.15

Key Findings

- PPO demonstrated strong robustness to Gaussian sensor noise, maintaining stable rewards across varying noise levels.
- Training under noisy conditions reduced collision rates from 10% to 0%, indicating improved policy generalization.
- Partial observability caused greater performance degradation than additive sensor noise, highlighting the challenges of missing information.
- Recurrent PPO introduced temporal memory for handling partial observability, but under the current training budget it did not consistently outperform the standard PPO baseline.

Conclusion

This study evaluated the robustness of PPO-based autonomous driving agents under sensor noise and partial observability in Highway-Env. Results show that standard PPO is highly resilient to noisy observations, while recurrent architectures provide additional benefits when critical information is consistently occluded. These findings demonstrate the importance of robustness-focused training and memory-based policies for reliable autonomous driving in uncertain real-world environments.

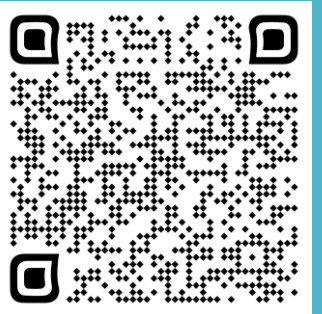
Resources



Project Page



Github



Drive